
2019-Ph-H1-Q14

Section: Particles and Waves

Topic: Radiation

Question:

A radioactive source has an initial activity of 800 Bq and a half-life of 10 hours. What is its activity after 30 hours?

Answer:

30 hours = 3 half-lives

Activity after 1: 800 → 400

After 2: 400 → 200

After 3: 200 → 100

Correct Option: C (100 Bq)

Guidance for Students:

Each half-life = divide by 2. Keep a tally or draw a table if unsure.

Revision Tips:

- Don't try to use equations — just halve repeatedly
 - Time passed ÷ half-life = number of half-lives
 - Always check if the time is an **exact multiple** of the half-life
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